Nirma University
Institute of Technology



Course Code :- 6CS105CC26

Course Name :- Computer Network

Practical - 5

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Implementation of a Network Servers(DHCP, FTP) using Cisco Packet Tracer

Aim

Implementation of Network Server Configurations (DHCP, FTP) using Cisco Packet Tracer.

Requirements

- Cisco Packet Tracer
- 3 PCs
- 1 Switch
- 1 DHCP Server
- 1 FTP Server
- Ethernet cables

IP Configuration

Device	IP Address
Default Gateway	198.126.0.1
DHCP Server	198.126.0.2
FTP Server	198.126.0.3

Subnet Mask: 255.255.255.0

DNS Server: 8.8.8.8

Procedure

1. Network Topology Design:

• Deploy two generic servers (designated as DHCP Server and FTP Server) and a 2960-24TT Switch.

• Deploy three end-user PCs (PC0, PC1, and PC2).

• Interconnect all devices using Copper Straight-Through cables via their FastEthernet interfaces.

2. DHCP Server Configuration:

• Access the DHCP Server, navigate to the Services tab, and select DHCP.

• Turn the DHCP Service On.

• Configure the pool parameters (Pool Name: serverPool, Default Gateway: 192.168.1.1, DNS Server: 8.8.8.8).

• Define the IP distribution parameters: set the Start IP Address to 192.168.1.67 and limit the Maximum Number of Users to 25.

• Click Save to apply the configuration.

3. FTP Server Configuration:

• Access the FTP Server, navigate to the Services tab, and select FTP.

• Ensure the FTP Service is turned On.

• Under User Setup, create authorized user profiles with specific permissions (e.g., username cheel with password strongPassword, and username student with password 123).

• Assign appropriate read, write, delete, rename, and listing permissions (RWDNL or RWL) for each user profile and click Add.

4. Verification and Testing:

• Allow client PCs to automatically request and receive an IP configuration from the DHCP pool.

• Access the Command Prompt on a client PC (e.g., PC0).

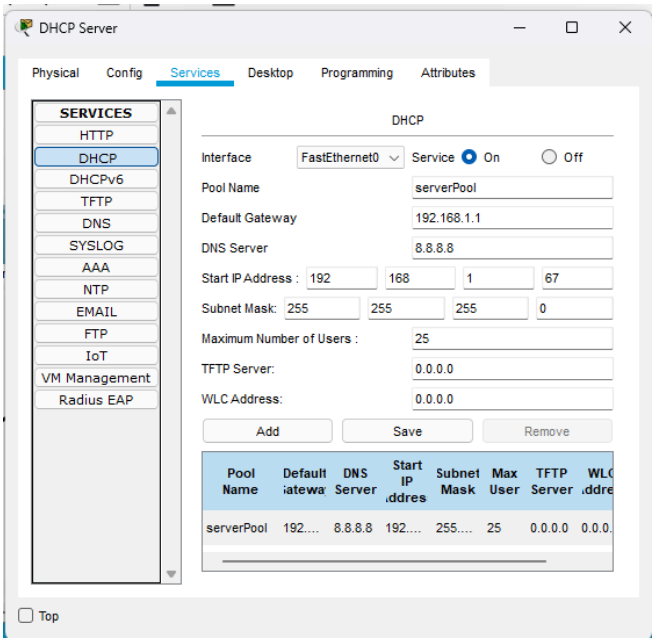
• Authenticate and connect to the FTP Server using its static IP address (192.168.1.3).

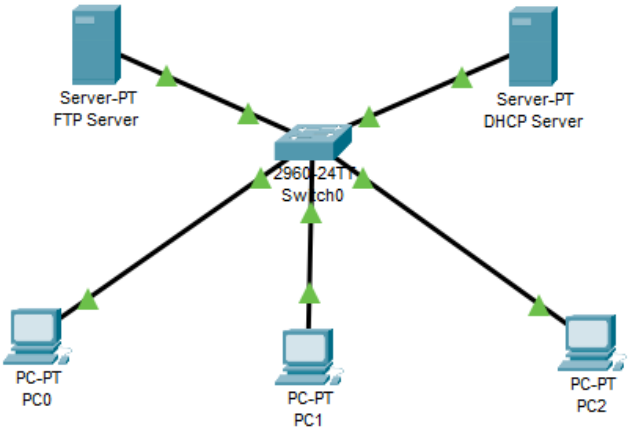
• Perform file manipulation actions, such as uploading local files to the server using the put command, and check the server's updated file directory.

Implementation

• DHCP Server Deployment:

The DHCP pool configuration successfully reserves initial subnet blocks and automates network provisioning. Client machines connecting to this network are dynamically assigned unique IP addresses beginning from 192.168.1.67 with a matching subnet mask of 255.255.255.0.

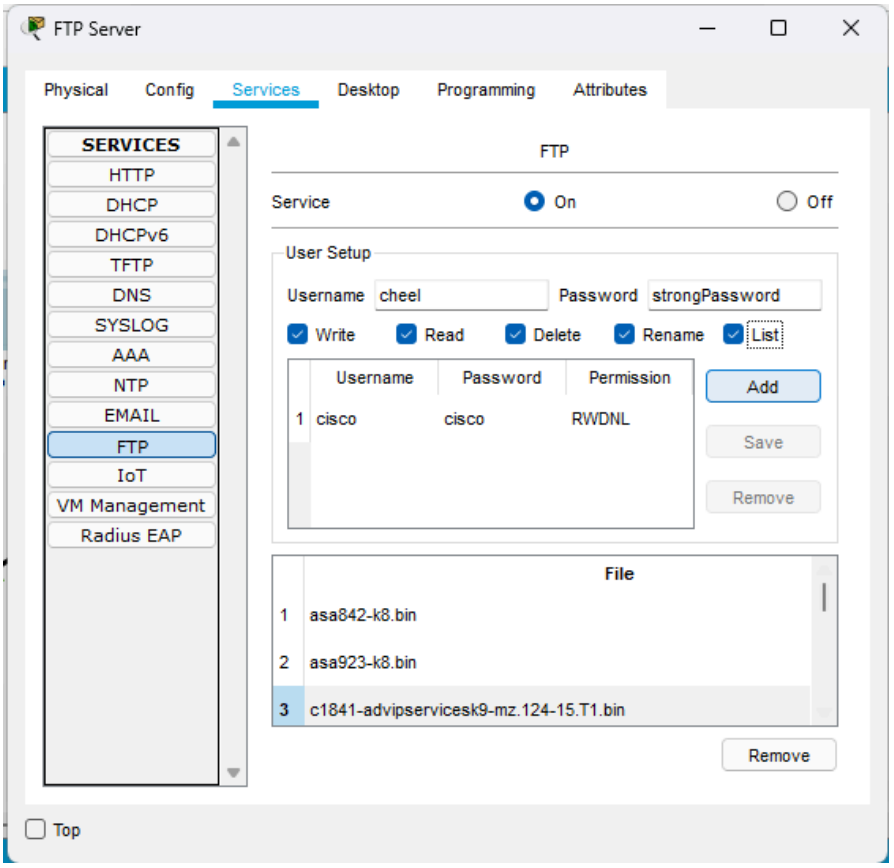


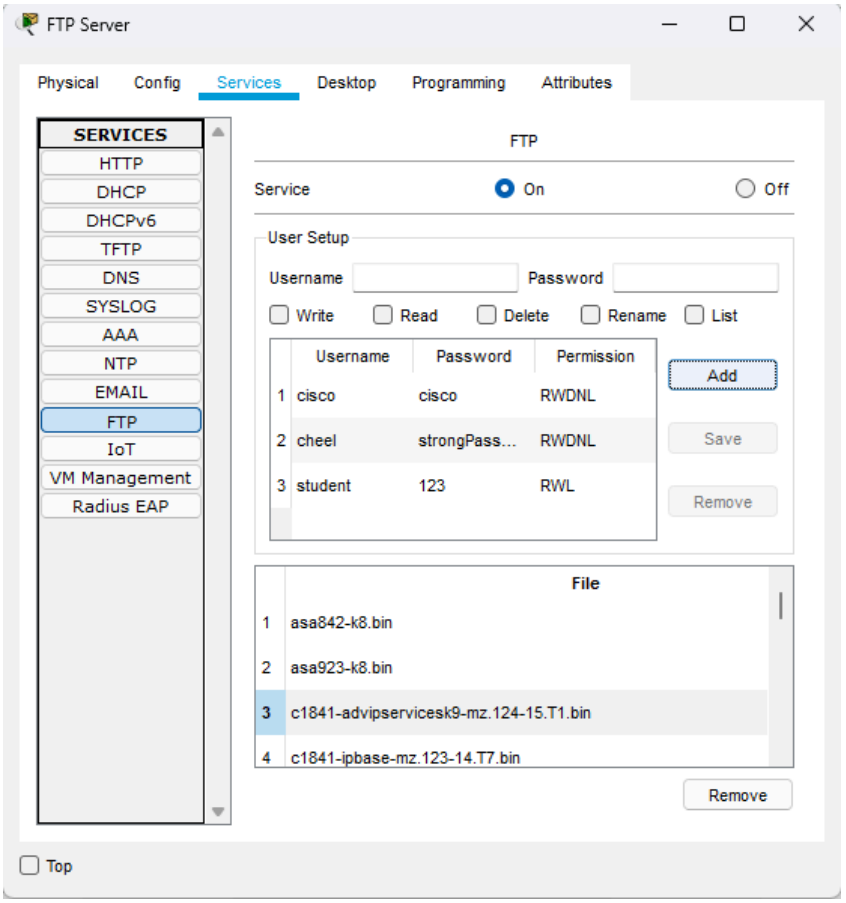


• **FTP Service Setup:**

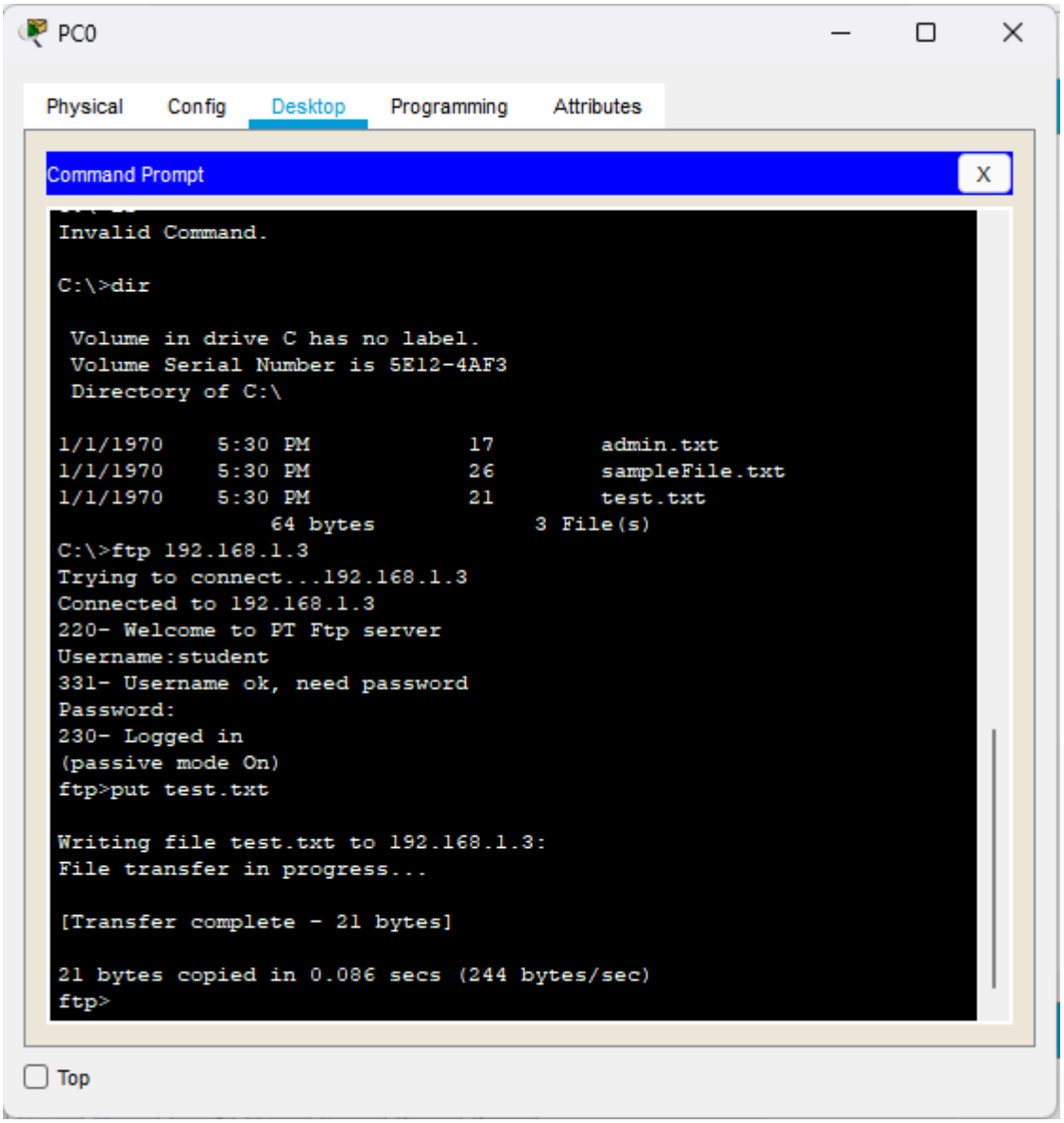
The FTP Server successfully hosts user databases. As shown in the configuration screens:

- ❖ User cheel has full rights (RWDNL).
- ❖ User student is configured with Read, Write, and List rights (RWL).





Result



From the terminal interface of PC0, a remote session was initiated via the command:bash

```
ftp 192.168.1.3
```

The user successfully authenticated using the credentials for student. The client then verified local files via dir and uploaded test.txt to the server storage array via the command:bash

```
ftp> put test.txt
```

Use code with caution.

The file transfer was confirmed complete with 21 bytes copied in 0.086 seconds.

Conclusion

The implementation of centralized network servers was successfully completed and verified. The DHCP server effectively eliminates the administrative overhead of manual IP configuration across multiple client machines, preventing address conflicts. Additionally, the FTP server successfully established restricted, credential-based storage access, allowing secure and verified data transfers across the local network infrastructure.